Lesson notes · for you

Agent engineering: planning, multi-agent, evals and guardrails: unit check

AI agents · University · Unit check

Time	Materials
20 min	Paper and a pen. No device needed.

Learning goals

- Plans and memory
- Several agents, one job
- Measuring an agent
- Fine-tuning and guardrails

Activity: Activity: Agent design review

1. Warm-up (3 min): Write one idea from each of the four lessons, in your own words.
2. Design (8 min): Sketch an agent for a real task at your university, such as answering course questions or booking rooms: its steps, what it keeps in its window, its roles and its tools.
3. Review (7 min): Swap with a partner. Check their design against the unit: where would errors carry forward, what could fall out of memory, how would they test it, and which actions are guarded in code?
4. Share (2 min): Each pair shares the weakest point they found and how they would fix it.

Use made-up names and data in every design. The point is the review, not a perfect agent.

Discussion

- Which idea from this unit changes how you would build an agent?
- What would convince you that an agent is ready for real users?
- Which guard would you add first to an AI tool you already use?

Check yourself

- My most useful idea from this unit + the project I will use it in.

Try it this week
Pick the habit from this unit you need most (checking every step, pinning key facts, a fixed test set or exact approval in code) and use it on a real project this week.

Worksheet

Agent engineering: planning, multi-agent, evals and guardrails: unit check

AI agents · University · Unit check

Name _____ Class _____ Date _____

1 **Each step of a four-step agent is right 90% of the time. About how often is the whole task right?**

◯ About 90% ◯ About 66% ◯ About 36%

2 **A model's context window is its only memory within a conversation: what falls out of it is forgotten.**

◯ True ◯ False

3 **Which help an agent keep what matters as a conversation grows? Choose all that apply.**

☐ Short notes in place of finished exchanges ☐ Pinning the key facts, such as dates and places

☐ Dropping the oldest turns, with no notes ☐ Retrieving only the part of a document a step needs

☐ Keeping everything until the window overflows

4 **Match each role in a team agent to its job.**

Planner	• • • •	Approves actions and handles what's stopped
Checker	• • • •	Runs the tool and returns the result
Runner	• • • •	Tests the call against strict rules
Person	• • • •	Proposes the next tool call

5 **A test set scored by group. What does this program print?**

```
hits = {"en": 15, "ar": 14}
total = {"en": 16, "ar": 16}
print(hits["ar"] / total["ar"])
```

◯ 0.875 ◯ 14 ◯ 0.9375